

lecture

October 29, 2025

1 data representations

dataset of n items, let a user be a vector in a d dimensional space

preserve relationships of distances between the points.

Definition 1 (Distortion). given a metric $d(x, y)$, apply an embedding algorithm where

$$f : x \mapsto f(x) \quad (1)$$

and

$$f : d \mapsto f(d) \quad (2)$$

compare d and $f(d)$ with

$$d \leq f(d) \leq Dd \quad (3)$$

where D is the distortion of f

Theorem 1 (Bourgain's theorem). Every metric can be embedded with distortion $O(\log n)$ in ℓ_2

Theorem 2. for any $\varepsilon > 0, n \in \mathbb{R}^d$ for a projection f ,

$$(1 - \varepsilon)|x - y|^2 \leq |f(x) - f(y)|^2 \leq (1 + \varepsilon)|x - y|^2 \quad (4)$$

as long as the dimension of the projection is $O(\frac{\log n}{\varepsilon^2})$

2 monotone embeddings

cardinal vs ordinal

cardinality, values, numbers vs ordering

want to preserve comparisons between distances

“a is more similar to b than to c”

are there low dimensional ℓ_2 that respect ordinal comparisons (not the length)

Definition 2 (Ordinal Embedding).

let X be a metric space on points $\{1, 2, \dots, n\}$ with distance metric δ . we call $\phi: X \rightarrow \mathbb{R}^d$ an *Ordered Embedding* if

$$\forall x, y, z \in X, \delta(x, y) < \delta(y, z) \implies d(\phi(x), \phi(y)) < d(\phi(y), \phi(z)) \quad (5)$$

there are three types that make sense

1. $\binom{n}{3}$ triplets, (A, B, C) . (tis is the *dense regime*)
2. *terminal embeddings* have k special nodes $\{t_1, t_2, \dots, t_k\}$, if not terminal, like t_1, t_n then i dont care about order
3. *k-nnearest neighbor embeddings* where you only care about the close elements and the farthest elements ordering doesnt matter

the first and third options are still not fully understood, but the second is well understood

in general to preserve all $\binom{n}{3}$ it requires $d > \frac{n}{2}$ dimensions to embedd all triplets